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Ersatz für
GS 90010-2:2020-01

Types of surface protection for metallic materials

Metallic coatings and inorganic coating systems

Drawing components (structural components)

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Continued page 2 to 14

Contents

Foreword

The German version is binding.

This Group Standard has been coordinated with the responsible departments of the BMW Group. Author, approver and reviewer are stored in and can be retrieved from the TEREG system master data.

Amendments

The following amendments have been made to GS 90010-2:2020-01:

– In Table 5, the lawyer thickness and test duration NSS have been changed for surface protections ZnNi-p;
– Standard editorially revised.

Previous editions

GS 90010-2: 2020-01

1 Scope and purpose

This standard applies to metallic coatings and inorganic coating systems on metallic materials of structural components like e.g. chassis subframes with and without thread, brackets with or without threads, control arms, carriers, stabilizer links, wiper linkage, lock support, struts.

It is not applicable for:

– mechanical fasteners like standard parts or standard-similar drawing components, refer to GS 90010-1;
– metal parts with organic coating, see GS 90011;
– surfaces or components with mainly decorative purpose;
– for electroplated plastic parts refer to GS 97017.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

AA-0244	Paint compatibility test
DIN 50969-1	Prevention of hydrogen-induced brittle fracture of high-strength steel building elements; Part 1: Preventive measures
DIN 50969-2	Prevention of hydrogen-induced brittle fracture of high-strength steel building elements; Part 2: Tests
DIN EN 10346	Continuously hot-dip coated steel flat products for cold forming; Technical delivery conditions
DIN EN ISO 1463	Metallic and oxide coatings; layer thickness measurement; microscopic procedure
DIN EN ISO 2143	Anodizing of aluminum and its alloys; Estimation of loss of absorptive power of anodic oxidation coatings after sealing - Dye-spot test with prior acid treatment
DIN EN ISO 2819	Metallic coatings on metallic substrates - Electrodeposited and chemically deposited coatings - Review of methods available for testing adhesion
DIN EN ISO 9227	Corrosion tests in artificial atmospheres; salt spray tests
GS 90010-1	Types of surface protection for metallic materials; Standard parts, drawing parts
GS 90011	Coating of parts made of metallic materials by means of organic materials; requirements and tests
GS 91005-1	Technical drawings; drawing generation, design rules, agreements
GS 97017	Coatings on plastic parts; electroplated plastic parts; requirements, tests
GS 97102	PVD Coatings on stainless steel surfaces and chromed surfaces; requirements and tests
VDA 233-102	Cyclic corrosion testing of materials and components in automotive construction

3 General

3.1 Term Surface protection

Surface protection is achieved by chemical/electro-chemical or physical application of coatings or by changing the component surface into a corrosion or wear resistant condition.

Apart from possible decorative effects, the main objective of surface treatment is the achievement of corrosion protection adapted to stress and requirements.

3.2 Hydrogen-induced brittle failure

Hydrogen induced brittle failure may occur on high strength steel parts e.g. due to pickling or electroplating.

For parts with a tensile strength of $R_m \geq 1\,000$ MPa the surface and heat treatment processes shall comply with state-of-the-art technology for prevention of hydrogen induced brittle failures. This applies particularly for electro deposited coatings. Even with zinc flake coatings, the parts may not be pickled in acids without inhibitors during pretreatment. The time between pretreatment and coating shall be as short as possible.

Proof of process reliability can be provided by means of stress test according to DIN 50969-2. This shall be documented in the initial sample inspection report.

Electroplated zinc and zinc alloy coatings are not permitted for parts with a tensile strength of $R_m \geq 1\,200$ MPa.

In such cases

– dimensioning shall be such that materials with lower strength can be used alternately,

- a material shall be specified that is not susceptible to hydrogen induced brittle failure,
- a material shall be selected that is durable without surface treatment or
- a different coating process shall be selected in agreement with the specialist department for corrosion resistance.

Examples for these coating processes:

- special processes (ZNS2, ZNS3);
- mechanical plating;
- thermo-chemical treatment;
- organic coating.

These parts shall always be cleaned mechanically or caustically. If chemical cleaning is required for functional reasons, only acids with suitable inhibitors may be used for preparation/cleaning. If chemical cleaning is required due to functional reasons only acids with suitable inhibitors shall be used for the preparation/cleaning.

In case of exceptions, which are subject to agreement with material development, measures shall be taken in accordance with DIN 50969-1 to avoid hydrogen-induced brittle fractures. These shall be documented in the initial sample inspection report.

3.3 Layer thicknesses

The values specified in the tables are recommended minimum layer thicknesses. For the release, the fulfillment of the respective corrosion protection requirement is decisive. Compliance with the respective corrosion protection requirement is decisive for the release/acceptance.

During the coating process the manufacturer shall ensure that the finished dimension of the parts is within the permitted tolerances (see 7.4).

Processes for the large-scale application of zinc flake coatings often require compromises in the uniformity of coating thicknesses.

If, in justified cases, certain layer thicknesses may not be exceeded, e.g. in downstream processes such as welding, the maximum permissible layer thickness shall be specified in the drawing.

In case of components with additionally added metric threaded parts the layer thickness depends on the coating process and the geometry of the threaded component. Required rework shall be harmonized with "Corrosion Resistance" specialist department and be specified in the drawing.

3.4 Contact corrosion

In order to avoid contact corrosion of steel components joined with aluminum components, one the following coatings shall be applied to the steel component, depending on the application case: ZnNi-p-v, ZnNi-p-v SW or Zn-I-T.

If subsequent painting is planned, coatings without additional sealing shall be used. Phosphatability shall be ensured.

For the BIW also ZnNi-p and Zn-p are allowed.

In case of contact with magnesium, stainless steel and carbon fiber reinforced plastics (CFRP), the selection of the type of surface protection or the material is subject to agreement with the "Corrosion Resistance" specialist department.

3.5 Electrically conductive joints

For electrically conductive connections (e.g. cable shoe) SnZn 70/30 shall be preferred. Other coatings shall be harmonized with the respective specialist departments.

For applications where electrical current flows only coating systems without sealing shall be used.

3.6 Coloration

Coloring may vary considerable depending on the surface. Due to process variances and ageing processes the coloration of the metallic coatings listed in this standard may vary. Hence, they shall not be used for decorative surfaces.

In case of anodic layers without color specification the manufacturer shall implement a silver finish.

3.7 Thermal joining

Thermal joining after coating is not allowed.

Exceptions shall be harmonized with specialist department „Corrosion Resistance“.

4 Properties and short description

4.1 Pretreatment

4.1.1 General

4.1.1.1 Legal specifications

All coating systems and materials referring to this standard shall comply with the legal requirements of guideline 2000/53/EC (EU end-of-life vehicle directive).

4.1.1.2 Unclosed cavities

Components may exhibit geometrically voluminous and complex open cavities. The coating quality shall be ensured for the inner surfaces of the components as well. As a rule, areas coated with insufficient quality and/or uncoated areas are not allowed, this applies to all sub-processes of the coating process including all pre-treatment measures.

If coating of inside cavities is required, dip coating shall be applied for all process steps. All components shall be fully dipped. It shall be ensured that all components are ventilated sufficiently quickly and completely irrespective of their position in the dipping bath and that sufficient process time is ensured for all surfaces of inside cavities. It shall also be ensured that the components are drained as far as possible between the individual dipping baths. Sufficiently large drain openings shall be provided for this purpose.

4.1.2 Degreasing and pickling

In order to ensure adhesion of the coating, residues and depositions (e.g. release agent, scale, silicates rolled-in particles) from upstream manufacturing processes shall be removed completely using suitable processes (e.g. pickling on phosphor or sulfur basis) such that coating adhesion is not impaired.

Cleaning shall be carried out completely on the inside and outside of hollow bodies.

The pre-treatment with the subsequent metallic coating shall be carried out in a continuous process. Storage or transport of parts (leaving the plant premises) between the individual process steps are not permitted.

4.1.3 Phosphating (only for zinc flake coating)

For zinc flake coatings, a pre-treatment agreed with the specialist department "Corrosion Resistance" (e.g. zinc phosphating, Ti/Zr passivation) can be used to improve the adhesion and corrosion properties.

4.2 Zinc electroplating

In zinc electroplating the workpiece is immersed in a zinc electrolyte. The application of an electric current causes the coating to deposit on the workpiece (cathode).

The deposited layer thickness depends on the streamline density on the workpiece and the power and duration of the current flow. Electrically shielded workpiece areas (e.g. deep holes, gaps, cavities, corners with small radii) form a Faraday cage and are not coated or only coated to a minor extent. In these cases, additional measures such as geometry changes, the use of auxiliary anodes or a change in the coating system are necessary.

Zinc electroplate coatings (Zn-p) exhibit a good basic corrosion protection.

Zinc alloy coatings are used for high corrosion protection requirements. The most common zinc alloy coatings are zinc-iron (ZnFe) and zinc-nickel (ZnNi).

Zinc-iron coatings contain 0,3 to 1,0 % iron as an alloying element.

Zinc-nickel coatings contain 10 % to 15 % nickel.

Corrosion protection is improved by passivation (p) and the application of an additional sealing (v) (refer to Table 1).

Passivation layers are barrier layers which are usually deposited from Cr(III) process solutions, are chemically stable and delay corrosive attack of the substrate. The most common processes are transparent and thick film passivations. Passivations can be transparent or colored in black (SW) for optical reasons.

Additionally applied sealing (e.g. sealers) are used to improve the appearance, long-term stability and functional properties of the coating. After prior approval by specialist department Corrosion Resistance the sealing may also be dyed unless this impairs the corrosion protection properties.

For components which are subjected to a downstream paint process the use of any type of sealing is not allowed. When a passivation is applied, phosphatability shall be ensured.

Furthermore, a harmlessness or paint compatibility test according to AA-0244 shall be demonstrated for such components.

In the case of special demands on surface functionality, the probable use of a sealer and the type of sealer shall be agreed with specialist department Corrosion Resistance, as the range of possible surface modifications by sealing is wide.

Electroplating (galvanizing) produces atomic hydrogen, which can diffuse into the parent metal. For parts with a tensile strength of $R_m \geq 1200$ MPa electroplated zinc and zinc alloy coatings are not permitted , refer also to Section 3.2.

4.3 Zinc flake coatings (Zn-I-T)

Zinc flake coatings (Zn-I) predominantly contain zinc flakes and up to 10 % aluminium flakes or alloy flakes with aluminium and/or magnesium (particle size approx. 0,25 µm ± 0,10 µm) in an inorganic or organic suspension. These are stoved for hardening. The stoving temperatures (min./max.) and stoving temperatures specified by the chemical manufacturer shall be observed.

The liquid coating material can be applied using the following application technologies

- spraying,
- dip-spin coating, and
- dip-drawing.

To assure coating adhesion, residues and deposits (e.g. parting agents, scale, silicates, rolled-in particles) from previous manufacturing processes shall be completely removed by suitable methods (e.g. pickling, peening) so that they do not impair coating adhesion in any way.

In cured condition, these coating shall be made up of mainly metallic constituents.

As a prerequisite for a cathodic protection effect the coating has to be electrically conducting. To prevent high settling rates and to achieve a good barrier layer, the metal particles in the coatings shall be aligned mainly parallel to the surface.

Zinc flake coatings have a low ductility. Therefore, these coatings are not allowed for components that are deformed or pressed-in after the coating process.

Their advantage over zinc electroplating is, above all, their ease of application and their suitability for large or unevenly shaped components, components with cavities, cup-shaped component geometries as well as the possibility to use them for high-strength steels, provided the pretreatment is adapted accordingly (see 3.2).

For sufficient corrosion protection, zinc flake coatings (Zn-I) shall be coated with an additional top coat (T) (Zn-I-T).

For components that undergo surface hardening during manufacture, the baking temperature shall not exceed 220 °C.

Due to the low adhesion of the zinc flakes to each other and to the substrate, adhesion to zinc flake coated components is not permitted.

Subsequent organic coating (paint) may only be applied after consultation with the specialist department "Corrosion Resistance".

Due to the system, zinc flake coatings have a low resistance to stone chipping.

Table 1: Zinc electroplatings and zinc flake coatings

Coating	without add. protection	Passivation	Top coat	Passivation + Sealing
Zinc	-	**Zn-p**	-	**Zn-p-v**
Zinc-iron, black	-	-	-	**ZnFe-p-v SW**
Zinc-iron	-	-	-	**ZnFe-p-v**
Zinc flake	-	-	**Zn-I-T**	-
Zinc-nickel, black	-	-	-	**ZnNi-p-v SW**
Zinc-nickel	-	**ZnNi-p**	-	**ZnNi-p-v**

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4.4 Hot-dipped zinc coating

Hot-dipped zinc coating means the coating of steel parts with a solid metallic zinc coating by immersion in a melt of liquid zinc or zinc alloy. Depending on the process, a distinction is made between discontinuous batch zinc coating and continuous hot-dipped zinc coating.

The corrosion protection of the hot-dipped zinc-coated component depends on the thickness of the coating and the additional alloying of elements such as aluminium and magnesium.

It is also possible to increase the corrosion protection by passivation or the use of sealants.

4.4.1 Batch zinc coating (HTG, NTG, ZnAl5)

In batch zinc coating, the processed component is dipped into the liquid melt after pretreatment. Due to the discontinuous process control, zinc coating results in significantly higher layer thicknesses and improved corrosion protection.

Depending on the temperature of the zinc coating bath, a distinction is made between normal temperature electroplating NTE (coating temperature ≈ 450 °C) and high temperature electroplating HTE (coating temperature ≈ 600 °C). The addition of approx. 5 % aluminum and the use of an additional sealant enables a significant reduction in the thickness of the ZnAl5 system layer with equivalent corrosion resistance.

Due to the high temperature of the molten zinc (in HTE and NTE) and the high layer thickness of the coating, the distortion and possibly the loss of dimensional accuracy of the components (especially in threads etc.) shall be taken into account.

Mechanical reworking may be necessary in threads and holes of batch zinc-coated components.

Table 2: Batch zinc coating process

Coating	Process temperature °C	Layer thickness µm	Alloying element	Sealing
HTG	560 to 630	25 to 80	-	not needed
NTG	450	40 to 80	-	not needed
ZnAl5	420	6 to 25	5 % Al	needed

4.4.2 Continuous hot-dipped zinc coating

Continuous hot-dipped zinc coated qualities according to Table 3 may only be used in consultation with the "Corrosion Resistance" specialist department in selected operating ranges.

When the part is formed, the trim edges are left without coating and are only protected against corrosion to a minor extent through the remote effect of the zinc. For highly stressed areas, such surface qualities shall not be used by preference. Exceptions and any additional measures shall be agreed upon with the "Corrosion Resistance" specialist department.

Table 3: Continuous hot-dipped zinc coating process (specifications according to DIN EN 10346

Quality	Melt composition	Average sheet thickness	Surface treatment			
Z100	≥ 99 % Zn	7 µm	C	O	CO	S
ZA095	5 % Al, 95 % Zn	7 µm	C	O	CO	S
ZM60	Mg, Al total 1,5 to 8 %, remainder Zn	4,5 µm	C	O	CO	S
Z275	≥ 99 % Zn	20 µm	C	O	CO	S
ZA185	5 % Al, 95 % Zn	14 µm	C	O	CO	S
ZM100	Mg, Al total 1,5 to 8 %, remainder Zn	5,5 µm	C	O	CO	S

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Any use of continuous hot-dipped zinc coated strips and sheets as final corrosion protection surfaces in the vehicle exterior area shall be agreed upon with the "Corrosion Resistance" specialist department. In general, the minimum corrosion protection requirement is passing the test according to VDA 233-102, three cycles without base material corrosion.

4.5 Hot-dip aluminizing

Hot-dip aluminized fine sheet metal is coating in a continuous process in an aluminum bath of 8 to 11 % silicon, up to 3 % iron, and the remainder aluminum. Hot-dip aluminizing offers no cathodic protection against ferrous materials and instead serves as a barrier to the base material or provides thermal and heat protection. For this reason, hot-dip aluminized thin sheet shall only be used for heat protection panels near the engine.

Table 4: Hot-dip aluminizing process (specifications according to DIN EN 10346)

Quality	Melt composition	Average sheet thickness	Surface treatment		
AS120	8 to 11 % Si, max. 3 % Fe, Remainder Al	20 µm	C	O	CO

4.6 Nickel electroplatings (Ni)

The corrosion protection of deposited electroplated nickel coatings is strongly dependent on the applied coating thickness and its homogeneity. The corrosion protection can be improved by a targeted combination of further deposited coating systems, such as the Cu/Ni/Cr system.

Electroplated nickel coatings can be used to improve hardness, wear protection, corrosion resistance, mechanical loadability, and scaling resistance.

In the case of deposited electroplated nickel coatings, a distinction is made between microporous and microcracked systems. The system used shall be specified in the initial sample inspection report. A change shall be indicated in all cases.

Electroplated nickel coatings have a hardness of 200 HV up to 500 HV depending on the type of coating.

Electroplated nickel coating is not permissible for parts with a tensile strength of R_m ≥ 1 200 MPa. See also 3.2.

4.7 Chemical nickel coatings (NiP-H)

The corrosion protection of electroplated deposited nickel/nickel phosphor coatings is strongly dependent on the applied coating thickness and its homogeneity. They exhibit higher corrosion protection than electroplated deposited nickel coatings.

Due to the deposition process, this surface can be used especially for components that need to have an internal coating or a conformal layer.

Depending on the phosphorus content and heat treatment, these coating systems have an additional hardenability of 550 HV up to 1 000 HV.

4.8 Hard chromium electroplating (Cr-H)

Hard chromium plating is the term used to describe particularly thick chromium coatings (usually layer thicknesses of a few µm to 1 000 µm).

Hard chromium coatings are characterized by high wear resistance and hardness of 800 to 1 100 HV and high temperature resistance.

When heated, the hardness of the chromium plating is retained up to approx. 350 °C. Due to recrystallization processes in the metal, the hardness decreases continuously at temperatures > 350 °C.

Due to the micro-cracked structure of the chromium layer, the substrate cannot be protected from corrosive influences with thinner layers. Sufficient corrosion protection with layer thicknesses ≤ 20 µm is only achieved in combination with other intermediate layers, e.g. nickel or copper/nickel, or by constant oiling.

Thicker chromium plating (layer thickness ≥ 20 µm) usually provide very good corrosion protection even without additional intermediate layers.

Chromium electroplating is not permissible for parts with a tensile strength of $R_m \geq 1\ 200$ MPa. For more details, see 3.2.

4.9 Tin electroplating (Sn)

Tin coatings do not offer anodic protection for steel materials. On copper and copper alloys, tin has very good oxidation protection, depending on the homogeneity of the coating. This ensures the good electrical properties and solderability of copper and copper alloys.

Tin electroplatings are only suitable for use in areas with low corrosive stress, e.g. for electrical ground connections.

Tin electroplating is not permissible for parts with a tensile strength of $R_m \geq 1\ 200$ MPa. See also 3.2.

4.10 Tin-zinc alloy electroplatings (SnZn 70/30)

These are high alloy tin coatings containing 65 to 75 % tin and 25 to 35 % zinc. Tin-zinc alloy electroplatings are used when reliable contact and high electrical conductivity are required, e.g. for electrical ground connections.

This surface coating is not suitable for areas with high corrosive stress (e.g. underbody area). In this case the use of these surface coatings shall be agreed with the "Corrosion Resistance" specialist department.

Tin electroplating is not permissible for parts with a tensile strength of $R_m \geq 1\ 200$ MPa. See also 3.2.

4.11 Anodizing layers

Anodizing layers are created by anodic oxidation of aluminum materials in an electrolyte. Properties such as corrosion protection and hardness can be specifically adapted by selecting the layer thickness and subsequent re-compacting.

4.11.1 Technical anodizing layers (AlO-T)

For technical applications, such as corrosion protection, the layers have a thickness of approx. 5 to 20 µm. Subsequent re-compaction improves the corrosion protection properties.

4.11.2 Hard anodizing layers (AlO-H)

Hard anodizing layers are used for wear and corrosion protection. These have layer thicknesses of a few µm to approx. 1 000 µm and are usually specified as drawing entries. Subsequent re-compaction reduces the wear protection properties and increases the corrosion protection properties.

Depending on the alloy used, hardnesses from 300 to 450 HV can be achieved.

5 Requirements

5.1 Corrosion protection requirements

Table 5: Functional coatings on steel

Coating metal	BMW code	Appearance	Layer thickness µm	Temperature resistance °C	Tests according to section 7.6			Coating or remark
					NSS h	VDA	TKW	
	-	-	min.	max.	-	-	-	-
Zinc	Body shop							
	Zn-p	silver	5[3]	-	240[1]	-	-	VDA 233-101
	ZnNi-p	silver	8[3]	-	720[1]	-	-	VDA 233-101
	Passenger compartment, luggage compartment							
	Zn-p-v[2]	silver	5[4]	120	360[1]	-	-	VDA 233-101
	ZnFe-p-v SW[2]	black	5[4]	150	120[5,6] 480[1,6]	-	-	VDA 233-101
	ZnFe-p-v[2]	silver	5[4]	150	480[1,6]	-	-	VDA 233-101
	Z100[11]	silver	5	180	-	3 cycles[1]	-	DIN EN 10346
	ZA095[11]	silver	5	200	-	3 cycles[1]	-	DIN EN 10346
	ZM60[11]	silver	4	180	-	3 cycles[1]	-	DIN EN 10346
	Vehicle exterior, underside, engine compartment							
	HTG	gray	> 25	180	-	3 cycles[1]	1 cycle[1,10]	DIN EN ISO 1461
	NTG	silver	> 40	180	-	3 cycles[1]	1 cycle[1,10]	DIN EN ISO 1461
	ZnAl5-v	silver	6 to 25	200	-	3 cycles[1]	1 cycle[1,10]	-
	ZnNi-p	silver	8	180	720[1,6]	6 cycles[1,9]	≥ 130 °C 1 cycle[1,10]	DIN EN ISO 19598
	ZnNi-p-v	silver	8[4]	180	720[1,6]	6 cycles[1,9]	≥ 130 °C 1 cycle[1,10]	DIN EN ISO 19598
	ZnNi-p-v-SW	black	8[4]	180	240[5,6] 720[1,6]	-	≥ 130 °C 1 cycle[1,10]	DIN EN ISO 19598
	Engine compartment only							
	Zn-I-T	silver	10	180	720[1,7]	-	≥ 130 °C 1 cycle[1,10]	-

Coating metal	BMW code	Appearance	Layer thickness µm	Temperature resistance °C	Tests according to section 7.6			Coating or remark
					NSS h	VDA	TKW	
Nickel	Exterior and interior							
	NiP-H	silver	20	240	240[1)]	-	-	DIN EN ISO 4527
Chromium	Cr-H (…)[8)]	silver	n.a.	350	n.a.	-	-	DIN EN ISO 6158
Tin	Sn	silver	8	-	144[1)]	-	-	-
Tin/zinc	SnZn 70/30[12)]	silver	10	150	480[1)]	-	-	-
Copper	Cu	copper	12	-	-	-	-	DIN EN ISO 1456

1) No base material corrosion (red rust) permissible.
2) Use in the direct field of vison only in agreement with the "Corrosion Resistance" specialist department.
3) Max. zinc coating 25 µm or max. 10 µm in weld areas.
4) plus sealing with system-dependent layer thicknesses between 0,5 µm and 2,0 µm.
5) No zinc corrosion (white rust) or visible changes permissible.
6) Test after preconditioning of the test specimens for 24 h at 120 °C.
7) If the components are used at ambient/component temperatures of > 130 °C, the test specimens shall be pre-conditioned for 96 h at 180 °C before the corrosion test.
8) The minimum layer thickness in µm shall be stated in the expression in parentheses. Hardness and corrosion resistance data as required.
9) Components which are mechanically formed after coating shall be put through a cyclic corrosion test according to VDA 233-102 on a further set of parts. Requirement no base material corrosion.
10) If the application temperature is permanently higher or equal to the specified temperature, an additional TKW according to 7.6.3 shall be performed.
11) Continuous hot-dipped zinc coating with greater coating weights is permissible in agreement with the Material Development specialist department.
12) Special coating, only valid for applications with separately stated conductivity requirements.

Table 6: Functional coatings on copper

Coating metal	Installation situation	BMW code	Appearance	Layer thicknes s µm	Max. temperature resistance °C	NSS test h
Tin	Exterior and interior	Sn	silver	8	-	144 1)
Nickel		Ni	silver	10	100	24
1) Test for base material corrosion.						

Table 7: Functional coatings on aluminum

Coating metal	Installation situation	BMW code	Appearance	Layer thickness µm	NSS test h
anodized	Exterior and interior	AlO-T	silver gray	15	720 2)
		AlO-H	various	n.V. 1)	240 2)
1) The hardness and minimum layer thickness in µm shall be stated in the drawing. 2) Test for base material corrosion.					

6 Designation and drawing entry

Only the BMW code may be used in the entry for surface protection in drawings and other documents.

Drawing entry in accordance with GS 91005–1.

Drawing entry for a low-alloy zinc electroplating, black passivate, sealed, code ZnFe-p-v SW:

GS 90010-2 - ZnFe-p-v SW

Non-standardized surface treatments shall be specified in the drawing after consultation with the Material Development specialist department, stating the exact coating designation and a suitable test.

7 Tests

7.1 Pretreatment

All part shall be cleaned or degreased prior to corrosion testing (e.g. using ethanol).

7.2 Initial sample inspection

The prerequisite for a release and delivery is the existence of a positive initial sample inspection report. In cases of dispute, the manufacturer is obligated, to present parts in good time and sufficient quantities (at least 5 parts).

7.3 Appearance and general requirements

The coatings shall not have any defects that may impair the appearance and/or the function. Parts from series supply shall correspond to the initial sample in terms of gloss, decorative appearance and corrosion resistance.

7.4 Layer thickness measurement

The measuring method for the layer thickness measurement can be selected freely. The resolution of the selected measuring method shall be adapted to the coating system to be measured. DIN EN ISO 1463 shall apply.

7.5 Adhesion

The adhesive strength of metallic coatings on metallic base materials shall be determined by the test methods listed in DIN EN ISO 2819.

The choice of adhesion test depends on the coating metal used and shall be made in consultation with the customer.

For zinc flake coatings, the adhesion test is carried out by means of an adhesive strip test.

In case of parts that are formed after coating, the protective coatings shall be such that they are not separated from the basic material or from each other and that they do not crack.

7.6 Corrosion tests

Parts from series deliveries shall fully comply with the initial sample in terms of corrosion resistance.

The parts are considered defective if base material/zinc corrosion occurs before the prescribed test period expires, or if the parts do not correspond to the initial sample in terms of appearance and corrosion resistance.

7.6.1 Neutral salt spray mist test (NSS)

In the neutral salt spray mist test NSS according to DIN EN ISO 9227, a neutral (pH 6,5 to 7,2) 5 % sodium chloride solution is constantly sprayed at T_{const}. = (35 ± 2) °C. The part to be tested is kept permanently moist in the chamber.

There is no direct correlation between resistance to the effects of salt spray mist and the long-term behavior of different coating or finishing systems in the vehicle, because the corrosion stress in the salt spray mist chamber differs significantly from that in practice.

Instead, the salt spray mist test serves as a test method for quality control of coatings and helps to identify coating defects and weak points in the coating.

Passivated or passivated and sealed parts put through a salt spray mist test NSS are heat treated before the test in order to account for the stress in operation, or possible impairments in case of any repainting.

7.6.2 Cyclic corrosion test (VDA)

The cyclic corrosion test according to VDA 233-102 is a corrosion test derived from real, corrosive climatic conditions.
This is a fast track test procedure with the aim of producing corrosion results that are as close as possible to practical application.
It consists of 3 different cycles, the salt spray, drying and cold cycles, in which the temperature (-15 to 50) °C and the humidity (50 to 100) % are uniformly changed in a ramp mode and combined with spraying salt spray mist.
Extreme ambient conditions in the vehicle such as a powerful incident airflow, high ambient temperature, etc. are not covered by the cyclic corrosion test according to VDA 233-102.

7.6.3 Temperature corrosion cycle test (TKW)

The temperature corrosion cycle test is a combination of heat treatment and a neutral salt spray mist test; it is used exclusively for components that are exposed to high ambient and object temperatures during driving operation.

7.6.3.1 Performing the test

Duration of one cycle = 1 week (= 5 quenching cycles + 163 h salt spray mist test according to DIN EN ISO 9227 NSS).
Sequence of one cycle:

a) Quenching cycle: 1 h thermal conditioning of the part for T_{duration}, where (T_{duration} is the actual application temperature of the part in operation; see also Table 5, index 10);
b) Quenching in saline solution (5 % NaCl, 20 °C), relocation to the salt spray mist chamber according to DIN EN ISO 9227, relocation time max. 10 s;
c) 23 h salt spray mist test according to DIN EN ISO 9227;
d) 4 repetitions of steps a) through c);
e) 48 h salt spray mist test according to DIN EN ISO 9227.

7.6.4 Other test procedures

Other test procedures and cycles shall be agreed upon with the specialist department "Corrosion Resistance" and stated in drawings and/or quality specifications.

7.7 Continuity in series

If details are not specified by the customer, or if the respective type of surface protection layer allows alternatives, the supplier may make a selection. The coater and the type of selected and sampled version shall be communicated to the customer e.g.

- pre- and post-treatment;
- in case of electroplatings, the type of alloy with information on concentration of alloying elements;
- type of coating material(s).

The sampled version of the surface protection is binding for the subsequent series and may only be changed with the approval of the customer. This also applies to the change of the coater and the coating process. This also applies to the change of the coater, the coating procedure and the coating process.

If initial sample inspection reports have been agreed upon, sample parts and initial sample inspection report are again required for all changes stated above.